Gautam Gupta

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Education

Indraprastha Institute of Information Technology Delhi — New Delhi, India
B.Tech in Computer Science And Engineering — *August 2023 - Expected July 2027*
Relevant Courses: Reinforcement Learning, Statistical & Bayesian ML, Data Structures & Algorithms, Computer Organization, Linear Algebra, Probability & Statistics, Multivariate Calculus, Introduction to Quantum Computing.

Technical Skills

Programming Languages: Python, C++, C, Java, CUDA, MATLAB
Machine Learning: PyTorch, TensorFlow, Hugging Face, Scikit-Learn, XGBoost
Development Tools: Git, VS Code, Jupyter, Claude-Code

Experience

AI Researcher — Remote
Mentors: Dr. Nils Lukas, Rushil Thareja (MBZUAI) — *August 2025 – November 2025*

- Developed a 'constitutional classifier' that learns auditable natural language privacy rules, achieving SOTA performance on Legal, Healthcare, and Finance documents without fine-tuning.
- Engineered the development of the Constitutional Tagger API service and deployed at website.

AI Researcher
Mentor: Rushil Thareja — *June 2025 – Aug 2025*

- Developed a demonstration for the paper DP-Fusion, deployed at website.
- Built Alpine Chat, a locally deployed, privacy-first chat interface with PII removal.

Undergraduate Researcher
Mentor: Dr. Rajiv Ratan (IIIT Delhi) — *October 2024 – April 2025*

- Architected a novel LLM-driven patch-generation algorithm to mitigate alignment-faking.
- Integrated Retrieval-Augmented Generation (RAG) with knowledge-graph embeddings to boost code patch accuracy.

Publications

MAC: Multi-Agent Constitution Learning
[ICML 2026 Submission]

- Proposed MAC, a multi-agent constitutional learning algorithm that automates the learning of transparent, high-performing constitutions by iteratively refining rules, outperforming recent prompt optimization methods by over 50% without requiring parameter updates.

Better and Worse with Scale: How Contextual Entrainment Diverges with Model Size
ACL 2026

- Formalized scaling laws for contextual entrainment, identifying that larger models become more resistant to misinformation but more prone to mechanical copying.

Demo: Sanitizing Medical Documents with Differential Privacy using LLMs
[GenAI4Health @NeurIPS]

- Introduced "constitutional classifiers" to accurately tag private data by learning auditable natural language rules, eliminating the need for costly fine-tuning.

Projects

Aerial Mapping Drone | *Python, PyTorch, CUDA* | GitHub April 2024 – September 2024

- Developed a unified 3D Gaussian Splatting pipeline and CUDA-accelerated VR renderer for real-time video to 3d reconstruction.

AI for Architects | *React, Python, Backend* | GitHub May 2025 – July 2025

- Developed a platform enabling architecture firms to rapidly generate on-brief sample images for client presentations.

AlphaOth Zero | *Python, PyTorch, RL* | GitHub May 2025 – June 2025

- Implemented AlphaGo Zero from scratch for Othello using self-play RL and Monte Carlo Tree Search (MCTS).

Super-Resolution Web-App | *Python, PyTorch, SRGAN* | GitHub August 2024 – September 2024

- Deployed an SRGAN-based web application allowing users to upscale and enhance low-resolution images in real-time.